

The answer key for the **Team** round is below.

p	Answer	p	Answer
1	4051	6	$\frac{47}{280}$
2	14283	7	$\frac{3+\sqrt{6}}{6}$
3	72	8	470
4	1170	9	1728
5	162	10	$\frac{99}{100}$

Problem 1. How many integers x satisfy $|x - 1434| \leq 2025$?

Proposed by: Taison Scofield

Answer: 4051

Solution: This inequality is equivalent to $-2025 \leq x - 1434 \leq 2025 \implies -591 \leq x \leq 3459$. To find how many integers x satisfy this inequality, compute $3459 - (-591) + 1$ (to account for $x = 0$) = 4051.

Problem 2. Let x and y be natural numbers such that

$$\gcd(x, y) = 69 \quad \text{and} \quad \text{lcm}(x, y) = 207.$$

Determine xy .

Proposed by: Zack Berchenko

Answer: 14283

Solution: Observe that

$$xy = \gcd(x, y) \cdot \text{lcm}(x, y) = 69 \cdot 207 = 14283.$$

Problem 3. An integer is *fiveminded* if it is divisible by 5 but does not contain the digit 5. How many three-digit fiveminded integers are there?

Proposed by: Samuel Kretzschmar

Answer: 72

Solution: We consider each digit separately. The last digit must be 0 or 5 to be divisible by 5, so the last digit must be 0 (only 1 choice). The first digit can be anything other than 5 or 0, so there are 8 choices for the first digit. The second digit can be anything other than 5, so there are 9 choices. As these are all independent decisions, the total number of three-digit fiveminded integers is $1 \cdot 8 \cdot 9 = 72$.

Problem 4. The polynomial $x^3 - 22x^2 + 27x + c$ has two roots which differ by exactly 2. What is the maximum possible value of c ?

Proposed by: Jefferson Zhou

Answer: 1170

Solution: Let the roots be $y - 1, y + 1, z$. Then by Vieta's, $2y + z = 22$, and $y^2 - 1 + yz + y + yz - y = y^2 + 2yz - 1 = 27$, so $y^2 + 2yz = 28$. Observe that $4y^2 + 2yz = 44y$, so $3y^2 = 44y - 28$, and $3y^2 - 44y + 28 = (3y - 2)(y - 14) = 0$. As such, $y = 14$, or $y = \frac{2}{3}$. As such, $(y, z) = (14, -6)$ or $(y, z) = \left(\frac{2}{3}, \frac{62}{3}\right)$.

Now, by Vieta's, $c = -(y^2 - 1)z$. If $(y, z) = (14, -6)$, then this equals $195 \cdot 6 = 1170$, and if $(y, z) = \left(\frac{2}{3}, \frac{62}{3}\right)$, then this equals $\frac{5}{9} \cdot \frac{62}{3} = \frac{310}{27}$, which is clearly smaller. Thus, the answer is 1170.

Problem 5. Jeff writes down all of the permutations of the digits of 14341434 on the board. He calls a permutation *funny* if 434 is a substring of the permutation. (A substring must have all of its characters consecutive and in order). How many funny permutations did Jeff write on the whiteboard?

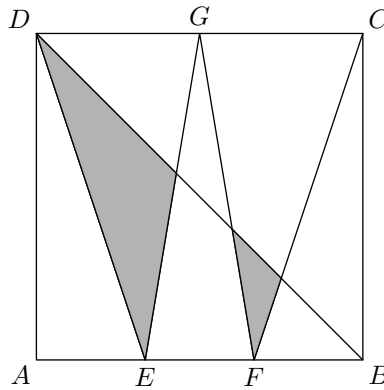
Proposed by: Jefferson Zhou

Answer: 162

Solution: We will first treat the first 434 as a single string, and find the number of permutations of 1(434)1434. This is $\binom{6}{2}\binom{4}{2} \cdot 2 = 180$.

However, this counts permutations with two substrings as 434 twice, so we need to subtract the number of said permutations. If the permutation has a 43434, then there are $\binom{4}{2} \cdot 2 = 12$ possible permutations, and if there are two separate 434's, then there are $\binom{4}{2} = 6$ possible permutations. Thus, the answer is $180 - 12 - 6 = \boxed{162}$.

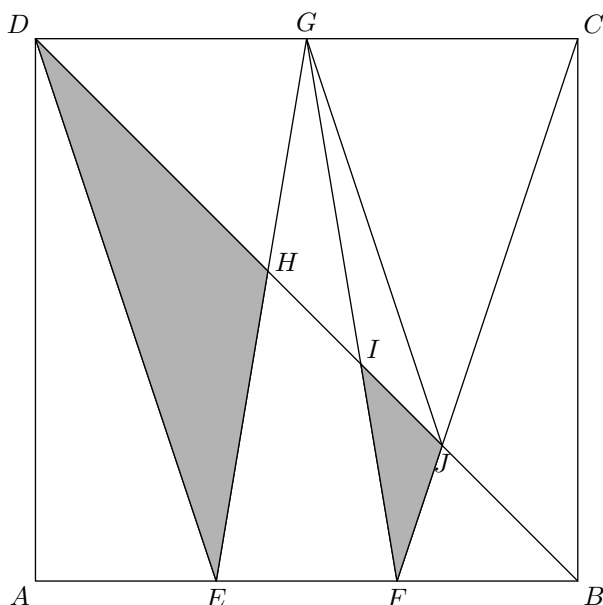
Problem 6. In square $ABCD$ below, $DG = GC$, and $AE = EF = FB$. What is the ratio of the shaded area to the area of the square?



Proposed by: Jefferson Zhou

Answer: $\frac{47}{280}$

Solution:



Let H be the intersection of GE and BD , let I be the intersection of GF and BD , and let J be the intersection of CF and BD .

Note that $\triangle DHG \sim \triangle BHE$, so $\frac{GH}{HE} = \frac{DG}{BE} = \frac{3}{4}$. Thus, $\frac{EH}{GE} = \frac{4}{7}$, so the area of DHE is $\frac{4}{7}$ of the area of DEG , which itself is $\frac{1}{4}$ of the area of the square. Thus, triangle DEH is $\frac{1}{7}$ of the area of the square.

Also, observe that $\triangle DIG \sim \triangle BIF$, so $\frac{GI}{IF} = \frac{DG}{BF} = \frac{3}{2}$, so $\frac{IF}{FG} = \frac{2}{5}$. Now, the area of IJF is $\frac{2}{5}$ of the area of FJG . Also, $\triangle DJC \sim \triangle BJF$, so $\frac{CJ}{JF} = \frac{DC}{BF} = 3$, so $\frac{JF}{CF} = \frac{1}{4}$, and so the area of GJF is $\frac{1}{4}$ of the area of GCF , which itself is $\frac{1}{4}$ of the area of the square. Thus, triangle IJF is $\frac{1}{40}$ of the area of the square.

Now, the total shaded area is $\frac{1}{7} + \frac{1}{40} = \boxed{\frac{47}{280}}$ of the area of the square.

Problem 7. Zack flips two unfair coins, each of which lands heads with probability $\frac{2}{3}$. Simultaneously, Jack flips two unfair coins, each of which lands heads with probability x , where $x > \frac{1}{2}$. A person wins if both of their coins land on the same face (either both heads or both tails), and the other person does not. If neither get matching faces on a flip, the process is repeated until someone does. If both get matching faces on the same flip, the game immediately ends and no one wins. Given that the probability that Jack wins is $\frac{2}{5}$, find the value of x .

Proposed by: Taison Scofield

Answer: $\boxed{\frac{3 + \sqrt{6}}{6}}$

Solution: The probability that Zack's two coins match (both heads or both tails) is $(\frac{2}{3})^2 + (\frac{1}{3})^2 = \frac{5}{9}$. The probability that Jack's two coins match is $x^2 + (1-x)^2 = 2x^2 - 2x + 1$. Let P_J be equal to this value. In each round:

- The probability that Zack matches and Jack does not: $\frac{5}{9} \cdot (1 - P_J)$,

- The probability that Jack matches and Zack does not: $P_J \cdot \frac{4}{9}$,
- The probability that both match (draw): $\frac{5}{9} \cdot P_J$,
- The probability that neither matches (repeat): $\frac{4}{9} \cdot (1 - P_J)$.

Because the game repeats until someone wins or there is a draw, the total probability that Jack wins is given by the sum of a geometric series. The chance Jack wins in one round is $a = P_J \cdot \frac{4}{9}$. The chance the game repeats (no winner) is $r = \frac{4}{9} \cdot (1 - P_J)$.

So total probability Jack wins is

$$\frac{a}{1 - r} = \frac{P_J \cdot \frac{4}{9}}{1 - \frac{4}{9}(1 - P_J)}.$$

The denominator is $1 - \frac{4}{9} + \frac{4}{9}P_J = \frac{5}{9} + \frac{4}{9}P_J$.

Therefore, Jack wins with probability

$$\frac{P_J \cdot \frac{4}{9}}{\frac{5}{9} + \frac{4}{9}P_J} = \frac{4P_J}{5 + 4P_J}.$$

Given the probability that Jack wins is $\frac{2}{5}$, we can solve $\frac{4P_J}{5+4P_J} = \frac{2}{5}$ by cross multiplying. This yields $P_J = \frac{5}{6}$. Now recall from the beginning that $P_J = 2x^2 - 2x + 1 = \frac{5}{6}$. We can multiply both sides by 6 and solve this with the quadratic formula, getting $x = \frac{3 \pm \sqrt{6}}{6}$.

Since $x > \frac{1}{2}$, the solution is $x = \frac{3 + \sqrt{6}}{6}$.

Problem 8. The sum of the positive divisors of a number x , including 1 and x , is 248. Find the sum of all possible values of x .

Proposed by: Taison Scofield

Answer: 470

Solution: We are given $\sigma(x) = 248$. Since $248 = 2^3 \cdot 31$, we try factorizations of $\sigma(x)$ using the formula $\sigma(p^a) = \frac{p^{a+1}-1}{p-1}$. The factors of 248 are 1, 2, 4, 8, 31, 62, 124, 248.

Testing values, we find $\sigma(2^4) = \sigma(5^2) = 31$, $\sigma(7^1) = 8$, $\sigma(3^1) = 4$, and $\sigma(61^1) = 62$. Thus possible factorizations are $31 \cdot 8 = 248$ and $62 \cdot 4 = 248$.

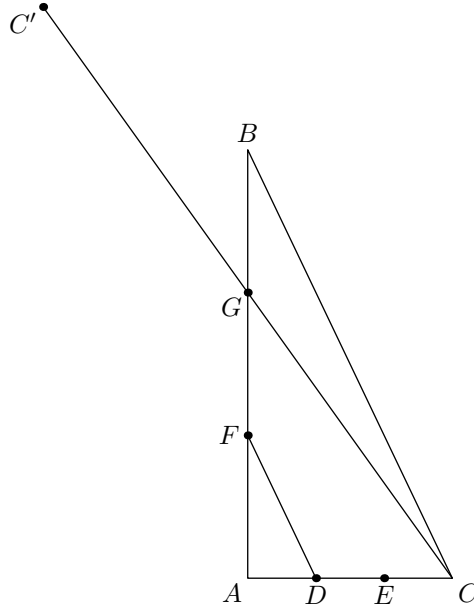
Therefore, the corresponding x are $2^4 \cdot 7 = 112$, $5^2 \cdot 7 = 175$, and $3 \cdot 61 = 183$. Their sum is 470.

Problem 9. In right triangle ABC , $AB = 24 + 6\sqrt{2}$, $AC = 24 - 6\sqrt{2}$, and $\angle BAC = 90^\circ$. Points D and E lie on segment AC such that $AD = DE = EC$ and D is between A and E , and points F and G lie on segment AB such that $AF = FG = GB$ and F lies between A and G . Let C' be the reflection of C across G . Line ℓ passes through C' and intersects rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ at P and Q , respectively. If $\angle QPA = \angle FDA$, what is the value of $C'P \cdot C'Q$?

Proposed by: Jefferson Zhou

Answer: 1728

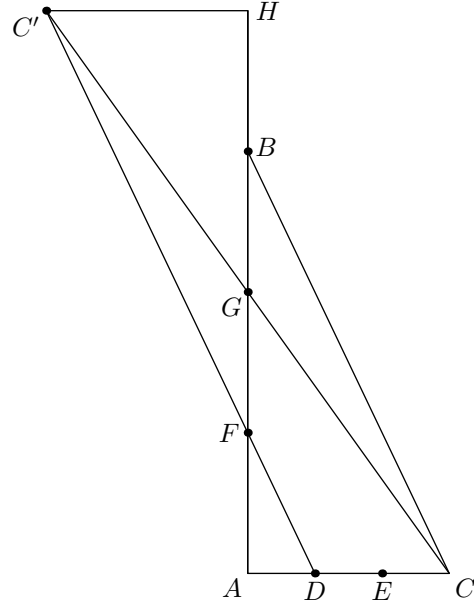
Solution:



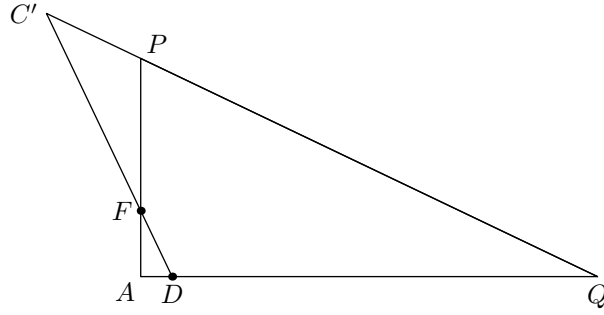
Firstly, observe that $AF = FG = GB = 8 + 2\sqrt{2}$, and $AD = DE = EC = 8 - 2\sqrt{2}$. Thus, by the Pythagorean theorem, $FD = \sqrt{(8 + 2\sqrt{2})^2 + (8 - 2\sqrt{2})^2} = 12$.

Claim. The points C' , F , and D are collinear.

Proof. Let H be the foot of the altitude from C' to AB .



Since C' is the reflection of C across G , we have $HG = AG = 2AF$ and $C'H = AC = 3AD$. Thus, $\frac{HF}{FA} = \frac{HG + FG}{FA} = \frac{3FA}{FA} = 3 = \frac{C'H}{AD}$, so since $\angle C'HF = 90^\circ = \angle DAF$, $\triangle DAF \sim \triangle C'HF$, so $\angle DFA = \angle C'FH$ and $C'F = 3FD = 36$. Now, since H , F , and A are collinear, we must have C' , F , and D collinear, as desired. $\square$



Now, observe that $\angle QPF + \angle FDQ = \angle QPA + 180^\circ - \angle FDA = 180^\circ$, so $PQFD$ is cyclic. Thus, by power of a point, $C'P \cdot C'Q = C'F \cdot C'D = 36(36+12) = 3 \cdot 4 \cdot 12^2 = 12^3 = \boxed{1728}$.

Problem 10. For all positive integers n , denote

$$f(n) = \frac{2n+1}{(n^2+2n+1)\sqrt[3]{n^2} + (n^2+n)\sqrt[3]{n^2+n} + n^2\sqrt[3]{n^2+2n+1}}.$$

Compute $f(1) + \cdots + f(999)$.

Proposed by: Jefferson Zhou

Answer: $\boxed{\frac{99}{100}}$

Solution: Note that

$$\begin{aligned} & \frac{2n+1}{(n^2+2n+1)\sqrt[3]{n^2} + (n^2+n)\sqrt[3]{n^2+n} + n^2\sqrt[3]{n^2+2n+1}} \\ &= \frac{2n+1}{\sqrt[3]{(n+1)^2n^2}(\sqrt[3]{(n+1)^4} + \sqrt[3]{n^2(n+1)^2} + \sqrt[3]{n^4})} \\ &= \frac{(2n+1)(\sqrt[3]{(n+1)^2} - \sqrt[3]{n^2})}{\sqrt[3]{(n+1)^2n^2}((\sqrt[3]{(n+1)^2})^3 - (\sqrt[3]{n^2})^3)} \\ &= \frac{(2n+1)(\sqrt[3]{(n+1)^2} - \sqrt[3]{n^2})}{(2n+1)\sqrt[3]{(n+1)^2n^2}} \\ &= \frac{\sqrt[3]{(n+1)^2} - \sqrt[3]{n^2}}{\sqrt[3]{(n+1)^2n^2}} \\ &= \frac{1}{\sqrt[3]{n^2}} - \frac{1}{\sqrt[3]{(n+1)^2}}. \end{aligned}$$

Thus,

$$\begin{aligned} & f(1) + f(2) + \cdots + f(999) \\ &= \left(\frac{1}{\sqrt[3]{1^2}} - \frac{1}{\sqrt[3]{2^2}} \right) + \left(\frac{1}{\sqrt[3]{2^2}} - \frac{1}{\sqrt[3]{3^2}} \right) + \cdots + \left(\frac{1}{\sqrt[3]{999^2}} - \frac{1}{\sqrt[3]{1000^2}} \right) \\ &= \frac{1}{\sqrt[3]{1^2}} - \frac{1}{\sqrt[3]{1000^2}} \\ &= 1 - \frac{1}{100} = \boxed{\frac{99}{100}}. \end{aligned}$$
